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Education

- **Korea University** Seoul, Korea
M.S. Computer Science and Engineering (GPA: 4.33/4.50) Sep. 2024 – Present
Advised by Professor Hyunwoo J. Kim, KAIST. (Expected Graduation: Feb. 2027)
B.S. Biomedical Engineering (GPA: 3.70/4.50) Mar. 2020 – Aug. 2024
Advised by Professor Dae Sung Yoon, Korea University. (Interdisciplinary Major in Artificial Intelligence)
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Publications

- [1] **RankAlloc: Model Merging with Spectral Adaptive Rank Allocation**
Chaehyeon Kim, Jihwan Park, Youngseo Cho, Gang Hyun Shin, Juyeon Ko, Hyunwoo J. Kim
Under Review
 - [2] **F4Splat: Feed-Forward Predictive Densification for Feed-Forward 3D Gaussian Splatting**
Injae Kim, Chaehyeon Kim, Minseong Bae, Minseok Joo, Hyunwoo J. Kim
Under Review
 - [3] **Study of Adversarial Attack and Defense Deep Learning Model for Autonomous Driving**
Chaehyeon Kim, Jinkyu Lee, Eun Jung, Jaeho Jung, Hyunjung Lee, Gyuyoung Lee
Annual Conference of KIPS (ACK 2022), vol. 29, no. 2, pp. 803–805, Nov. 2022. [Video]
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Research Interests

- Foundation model adaptation and capability composition, with a focus on model merging of task-specialized LLMs/MLLMs to preserve and compose their capabilities. Also interested in analyzing model behavior through weights, activations, representations, and outputs, with applications to complex multimodal tasks such as document understanding, where OCR, chart/table reasoning, visual grounding, and VQA are often required jointly.
 - Interested in post-training and alignment of foundation models, including supervised fine-tuning (SFT), preference optimization, RL-based training, and training data/objective design.
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Experience

- **MLV Lab** Seoul, Korea
M.S. Student at MLV Lab, Korea University Sep. 2024 – Present
 - Advised by Professor Hyunwoo J. Kim.
 - Conducting research on model merging for foundation models, with a focus on preserving and composing task-specific capabilities of LLMs/MLLMs.
 - Exploring model behavior analysis based on weights, activations, representations, and outputs, with interests in post-training and agentic workflows.
 - Contributing to a high-level 3D scene understanding and situation prediction project for urban drones, including 3D point cloud reconstruction, augmentation, and cross-modal data reconstruction.

- Participated in a medical image analysis project involving detection and segmentation of bacteria in Gram-stained microscopy images.

Undergraduate Research Intern at MLV Lab, Korea University

Nov. 2022 – Aug. 2024

- Advised by Professor Hyunwoo J. Kim.
- Conducted research on generative models and 3D vision, with a focus on diffusion models, point cloud processing, and 3D scene synthesis.
- Collected GTA-based simulated drone-view image data for a 3D urban scene understanding and situation prediction project.
- Conducted research on medical image analysis, including detection and segmentation of anterolateral thigh (ALT) perforators.

NEO Lab

Seoul, Korea

Undergraduate Research Intern at NEO Lab, Korea University

Mar. 2021 - Jun. 2021

- Advised by Professor Seungwoo Lee, Korea University.
- Studied DNA Neural Networks and conducted preliminary experiments.

Honors & Awards

- **Silver Prize**, University Division, 2023 Smart Maritime Logistics Competition, Ulsan Port Authority, Nov. 2023.
- **Honorable Mention**, 3rd L&P Cosmetic iPBL Idea Competition, Korea University Corporate-Industry Collaboration Center and L&P Cosmetic, Feb. 2023.
- **Top Excellence Award**, ICT Mentoring Academic Conference (ACK 2022), Korea Information Processing Society (KIPS), Dec. 2022. Project: "Study of Adversarial Attack and Defense Deep Learning Model for Autonomous Driving."
- **Grand Prize (Mayor of Busan Award)**, Digital Healthcare and Next-Generation Rehabilitation Welfare MEDICAL HACK 2021, Busan Metropolitan City, Aug. 2021.

Skills

- **Programming Skills:** Python, PyTorch, MATLAB, $\text{\LaTeX}$
- **Languages:** Korean (*native*), English (*intermediate*)

Research / Career Description

[Research] RankAlloc: Model Merging with Spectral Adaptive Rank Allocation

MLV Lab, KAIST

2026

- **Overview:** RankAlloc studies parameter-only model merging for combining multiple task-specialized models into a single model without additional training. The project addresses the rank allocation problem in subspace-based model merging: how to allocate limited representational capacity between common directions and task-specific residual directions. RankAlloc proposes a spectral adaptive rank allocation strategy that preserves a fixed top-ranked common subspace and adaptively assigns the remaining rank budget based on spectral energy.
- **Role:** Led the project end-to-end as the first author, including problem formulation, method design, implementation, experimental setup, baseline comparison, result analysis, figure/table preparation, and manuscript writing.
- **Methods / Tools:** Parameter-only model merging (Weight Averaging, Task Arithmetic [4], TIES-Merging [19], Consensus TA [17], TSV-M [1], Iso-Merging [12]), Singular Value Decomposition (SVD).
- **Key Contributions:**
 - Identified rank allocation across common and task-specific residual directions as an underexplored design factor in subspace-based model merging.
 - Proposed RankAlloc, a parameter-only model merging method that adaptively allocates rank budget using spectral energy of common and task-specific candidate directions.
 - Implemented the full RankAlloc pipeline, including common/task-specific subspace decomposition, candidate pool construction, spectral rank allocation, and merged update reconstruction.
 - Conducted extensive experiments across architectures and multi-task settings, comparing RankAlloc with representative weight-space and SVD-based merging baselines.
 - Analyzed task-wise heterogeneous rank demand and lower-ranked common-direction reallocation under constrained rank budgets.
- **Results:** Achieved the best performance among merging methods in the most constrained 20-task setting across ViT-B/32, ViT-B/16, and ViT-L/14, while remaining competitive in smaller 8-task and 14-task settings.
- **Status:** Under review.

[Research] F⁴Splat: Feed-Forward Predictive Densification for Feed-Forward 3D Gaussian Splatting

MLV Lab, KAIST

2025 – 2026

- **Overview:** F4Splat studies compact and efficient feed-forward 3D Gaussian Splatting for novel view synthesis. The project addresses the limitation of existing feed-forward 3DGS methods that allocate Gaussians uniformly or redundantly, and introduces predictive densification to adaptively allocate Gaussians according to scene complexity and multi-view overlap.
- **Role:** Contributed to method idea development, baseline reproduction, related work writing, figure design, and qualitative/quantitative analysis of F4Splat and baseline methods.
- **Methods / Tools:** 3D Gaussian Splatting (3DGS) [7], feed-forward 3DGS [20, 10, 3, 5], evaluation metrics (LPIPS, SSIM, PSNR).
- **Key Contributions:**
 - Reproduced baseline methods and established experimental settings for feed-forward 3D Gaussian Splatting.

- Contributed to the development of the main method idea, focusing on predictive densification and adaptive Gaussian allocation.
- Analyzed qualitative and quantitative results of F4Splat and baselines in terms of rendering quality, reconstruction fidelity, and Gaussian-count efficiency.
- Contributed to manuscript preparation and presentation of the method, including related work writing, experimental analysis, and visual design of the main figure.
- **Results:** Achieved superior novel-view synthesis performance compared to prior uncalibrated feed-forward methods, while using significantly fewer Gaussian primitives.
- **Status:** Under review.

[Project] High-Level 3D Scene Understanding & Situation Prediction for Urban Drones

MLV Lab, KAIST

2024 – Present

- **Overview:** This project aims to develop high-level 3D scene understanding and situation prediction technologies for urban drone environments.
- **Role:** Responsible for image and 3D data construction. For image data, contributed to simulated drone-view image collection, image-to-image translation, and human-object interaction (HOI) annotation. For 3D data, worked on point cloud augmentation, 2D-to-3D lifting, and cross-modal data reconstruction.
- **Methods / Tools:** GTAV simulation, Label Studio, image-to-image translation methods (CycleGAN [22], UNSB [8]), point cloud augmentation (PolarMix [18]), vision foundation models (VGGT [16], SAM [9], RLIPv2 [21]), cross-modal 3D reconstruction (MapAnything [6]), Open3D.
- **Key Contributions:**
 - Collected simulated drone-view image datasets in the GTAV environment by arranging objects according to predefined accident scenarios.
 - Applied synthetic-to-real image-to-image translation to reduce the domain gap between GTAV-style images and real-world images.
 - Annotated human-object interaction (HOI) labels for the collected image datasets using Label Studio.
 - Applied 3D point cloud augmentation methods to increase data diversity for urban drone scenarios.
 - Generated point clouds, segmentation masks, and HOI labels from image data using vision foundation models and explored 2D-to-3D lifting of the generated annotations.
 - Explored cross-modal 3D data reconstruction for connecting image-based observations with 3D scene representations.
- **Results / Status:** Ongoing project.

[Project] Study of Adversarial Attack & Defense Deep Learning Model for Autonomous Driving

ICT Mentoring (ACK 2022), Korea Information Processing Society (KIPS)

2022

- **Overview:** This project studied adversarial attack and defense methods for deep learning models in autonomous driving scenarios.
- **Role:** Led the project end-to-end as the first author, including problem formulation, experimental design, model implementation, attack/defense evaluation, result analysis, paper writing, and presentation.
- **Methods / Tools:** FGSM [2], PGD [11], MagNet [13], Defense-GAN [14], GTSRB dataset [15]

- **Key Contributions:**

- Led the end-to-end development of an adversarial robustness evaluation pipeline for GTSRB-based traffic sign recognition, including FGSM/PGD attack generation and defense evaluation.
- Proposed and implemented DefPCA, a PCA-based defense method that reconstructs adversarial images using principal components.
- Compared defense methods, including adversarial training, MagNet, Defense-GAN, and DefPCA, in terms of robustness and per-image defense time.
- Suggested practical deployment directions, including model lightweighting and selective defense using adversarial example detection.

- **Results / Status:** Published in Annual Conference of KIPS and received Top Excellence Award at ICT Mentoring Academic Conference (ACK 2022).

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